

QUIZ: CHAPTER 1 - ORGANIC CO

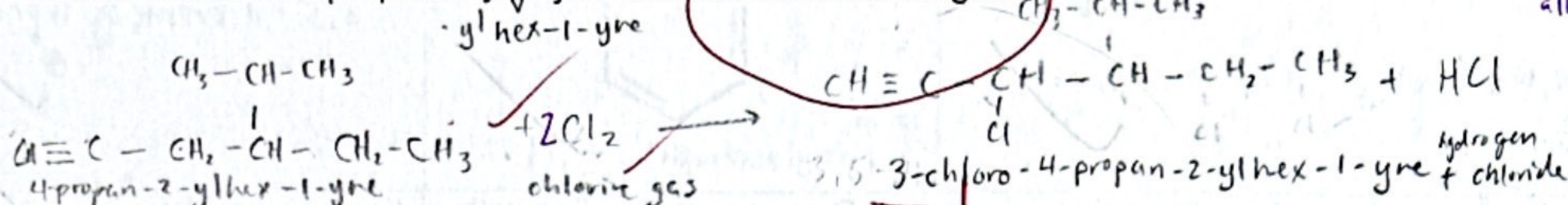
Knowledge /10 Inquiry 8.1

PART B: SHORT ANSWER QUESTIONS

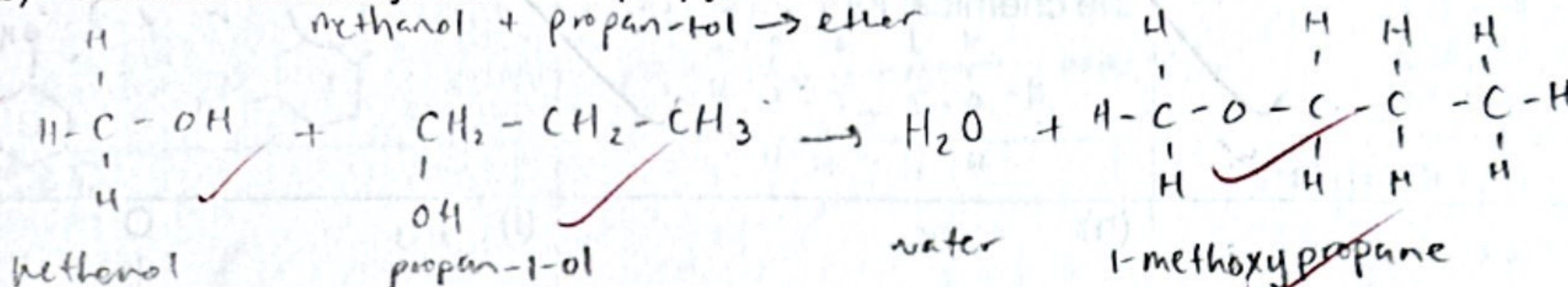
Inquiry (11 marks)

1. Write an equation, using structural formula or carbon skeleton, for each of the following reactions and **name the reactants and products**: (8 MARKS)

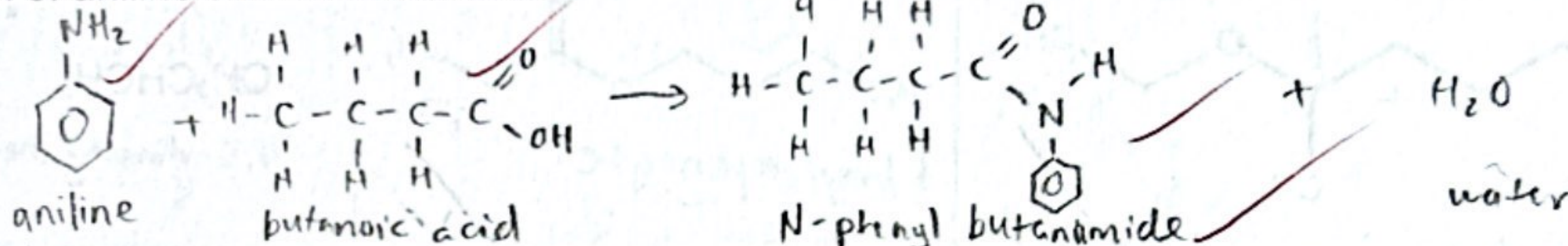
a) Reaction between 4-propan-2-yl-1-yne and excess chlorine gas



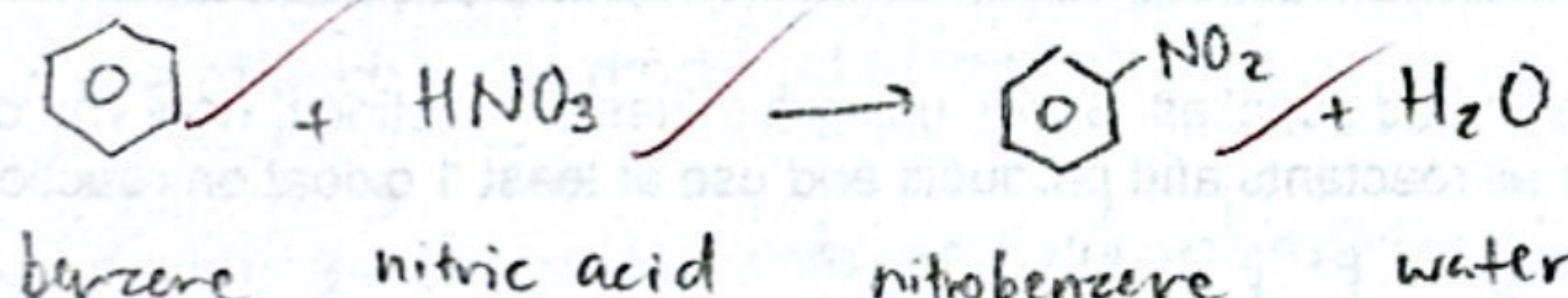
b) Reaction of methyl ether and propyl ether



c) Reaction of aniline and butanoic acid



d) Reaction between benzene and nitric acid



2. List the following 3 organic compounds in order of increasing boiling point in water. Explain in terms of **intermolecular forces**: dimethylether, ethane, ethanol (3 marks)

Lowest solubility	ethane	dimethylether	ethanol	Highest solubility
Explanation	The more polar something is, the higher solubility it has in water. Ethane is completely non-polar so it's last. Dimethyl ether has an oxygen bonded to 2 carbons, meaning that there are polar bonds with $\Delta\text{EN} = 1.0$. Ethanol has an oxygen bonded to a hydrogen, making it most polar at $\Delta\text{EN} = 1.4$.			

Communication (11 marks)

1. Provide 1 similarity and 1 difference between an ether and a carboxylic acid? (2 marks)

Similarity:

at least one

They both require an alcohol to be produced in a reaction.

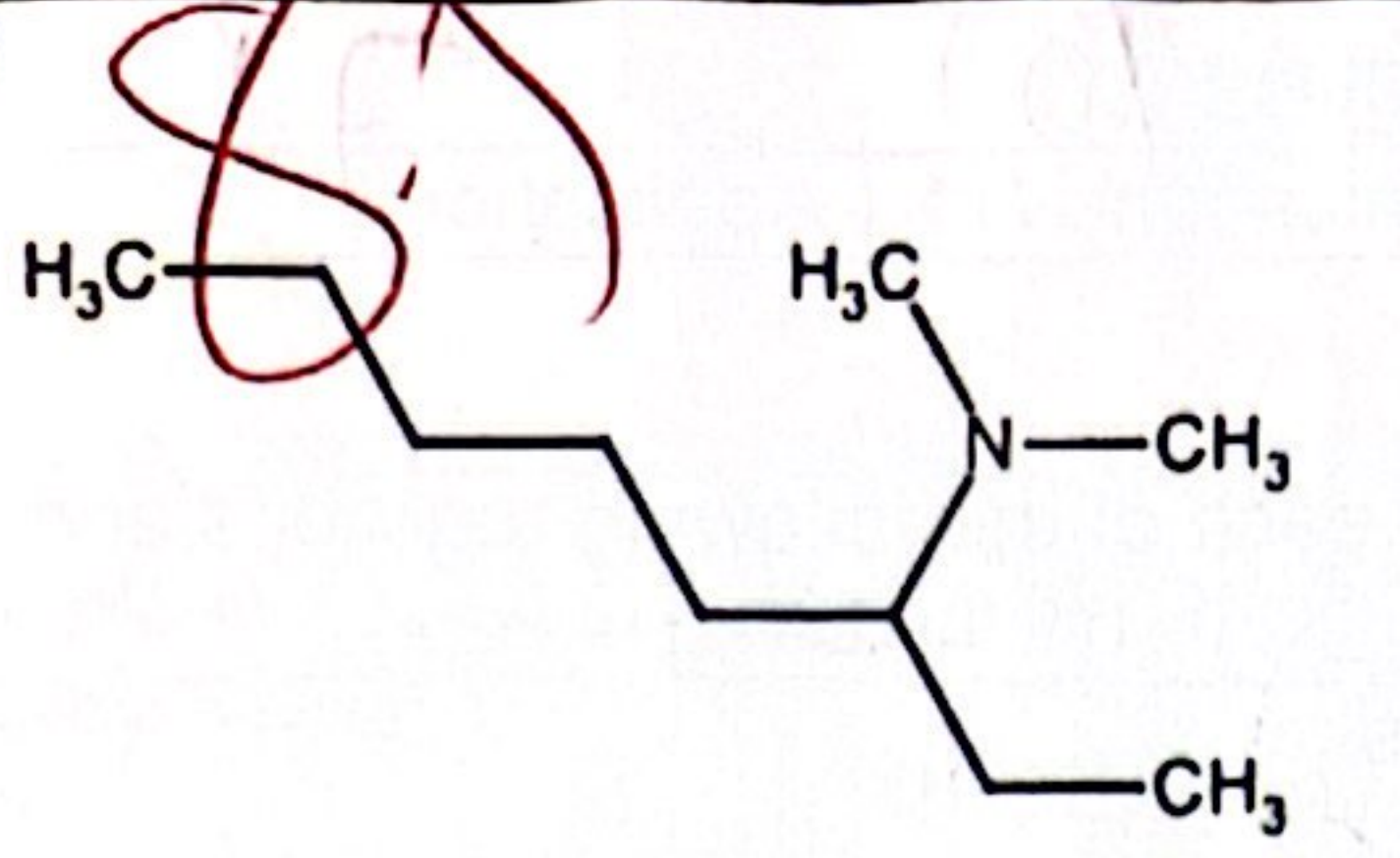
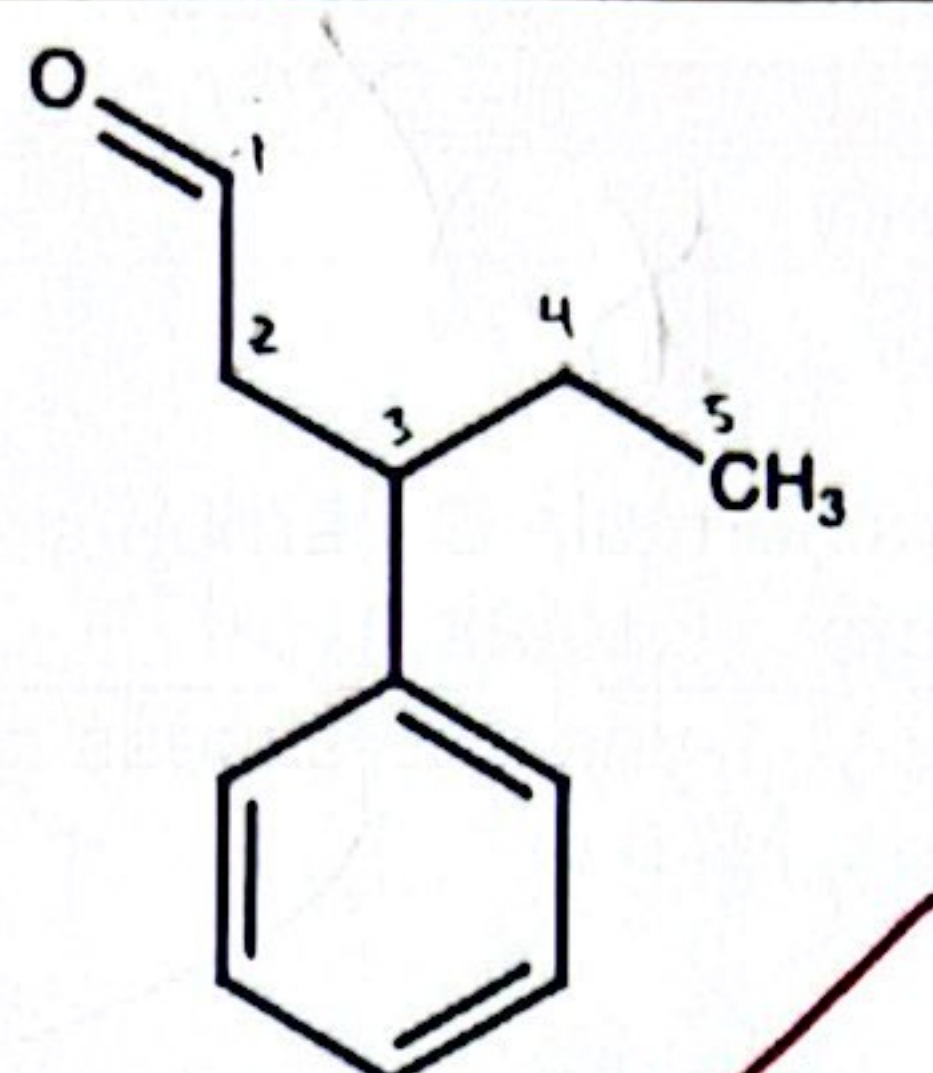
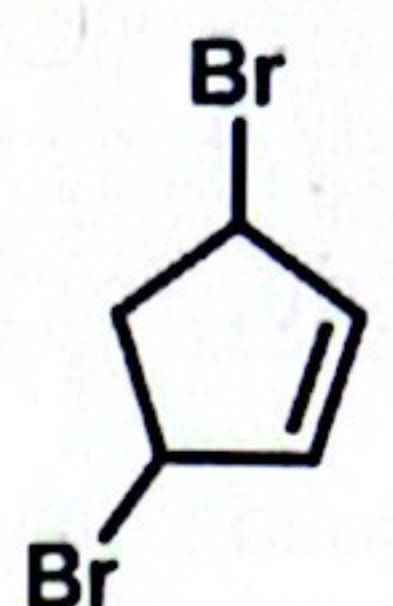
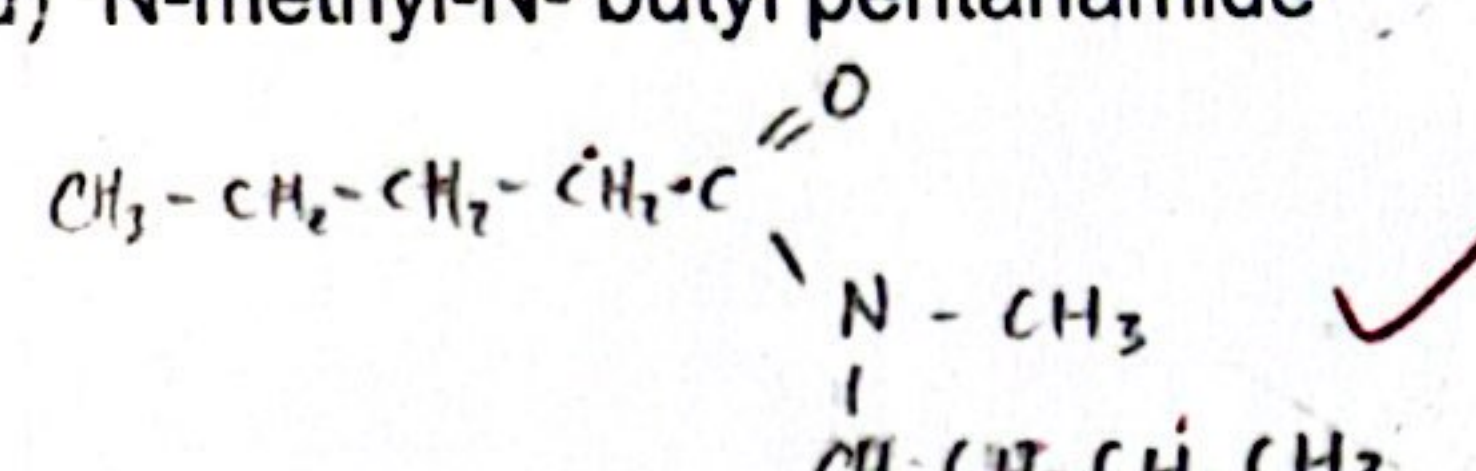
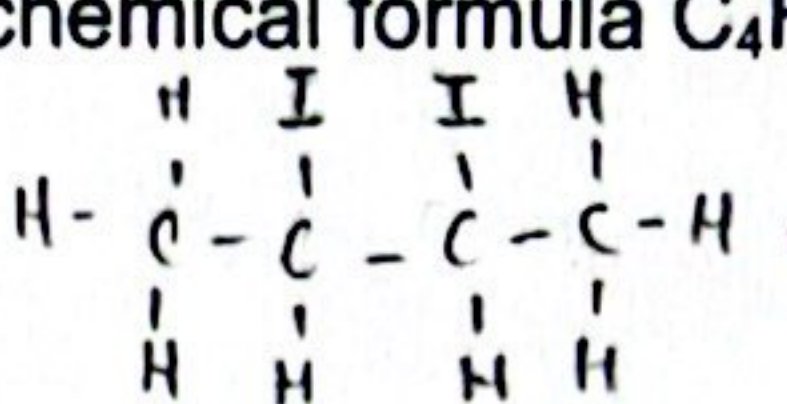
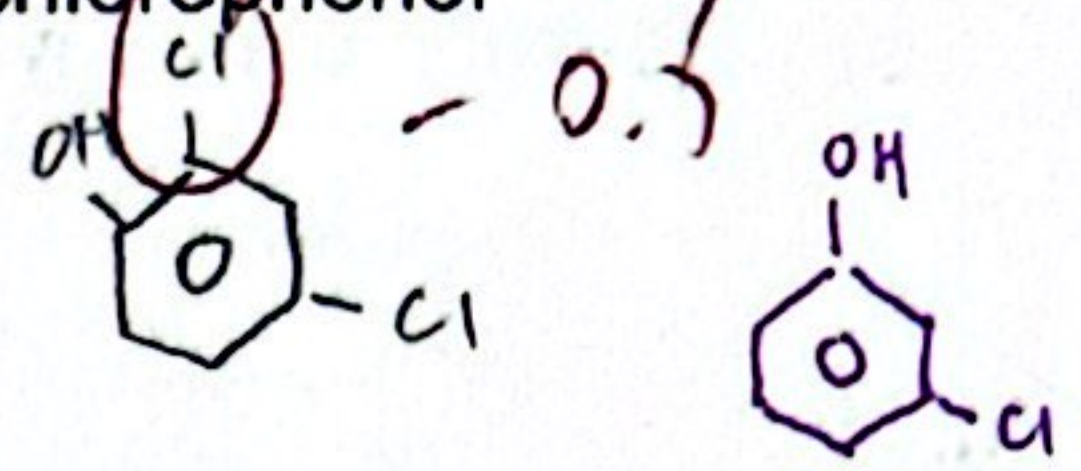
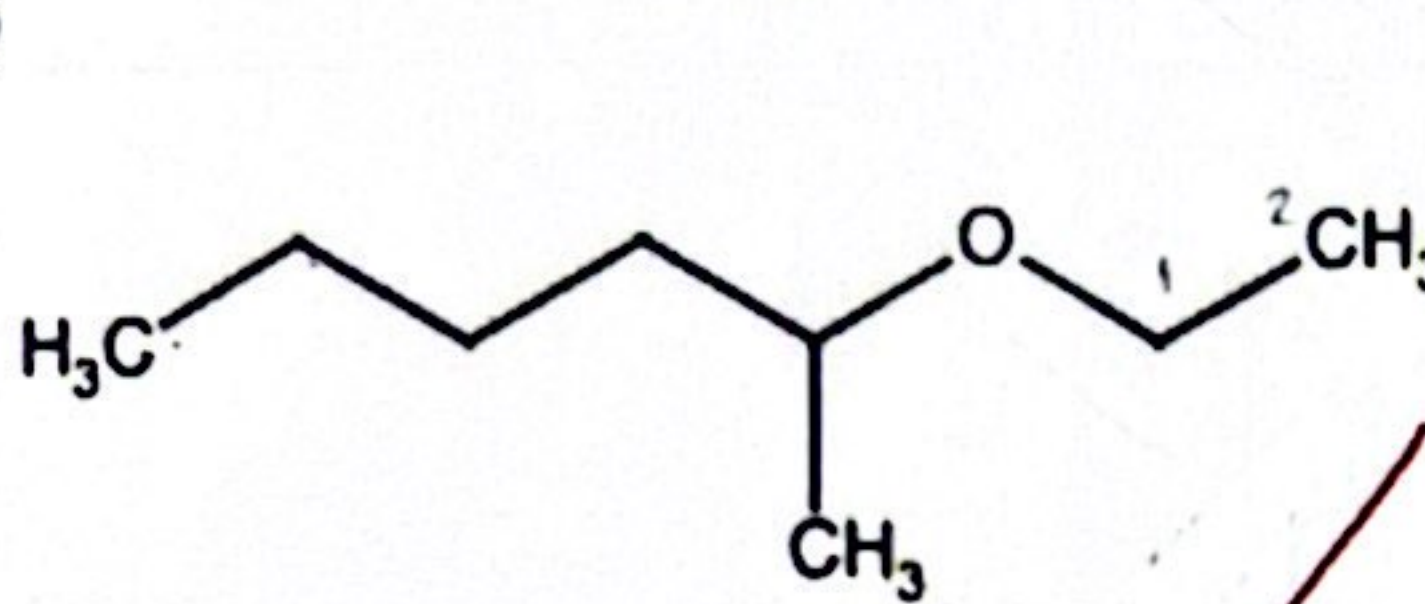
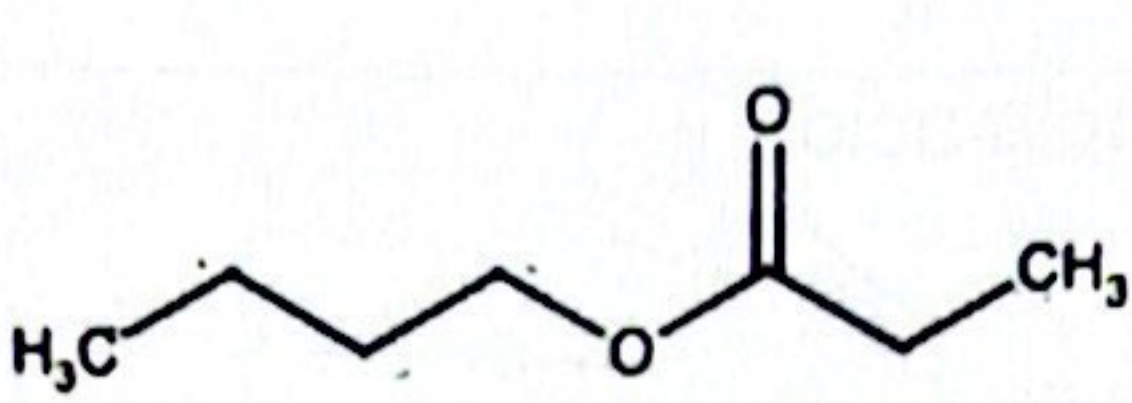
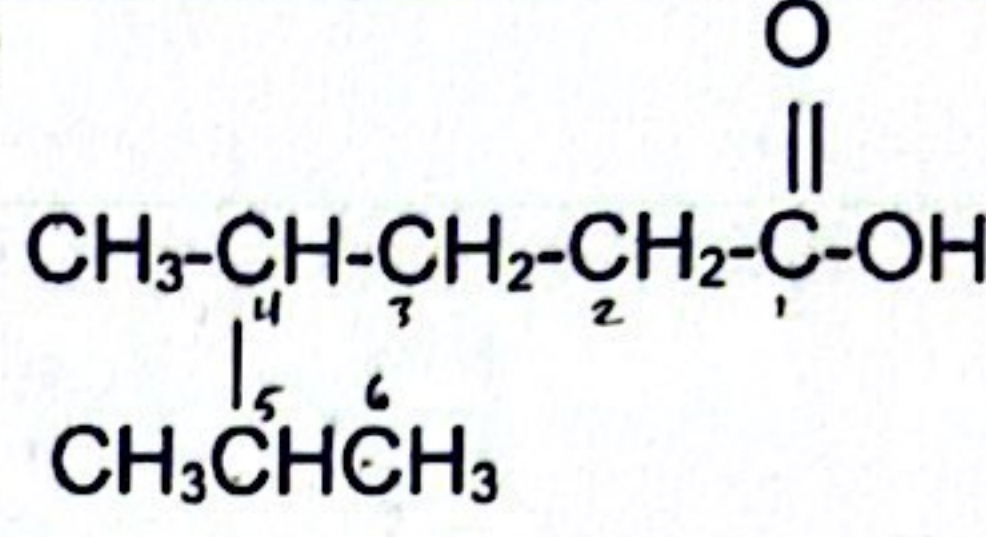
- have oxygen
- both polar

Difference:

carboxylic acids have hydrogen bonding, which ethers do not

- ether only has a double bonded O while carboxylic acids do

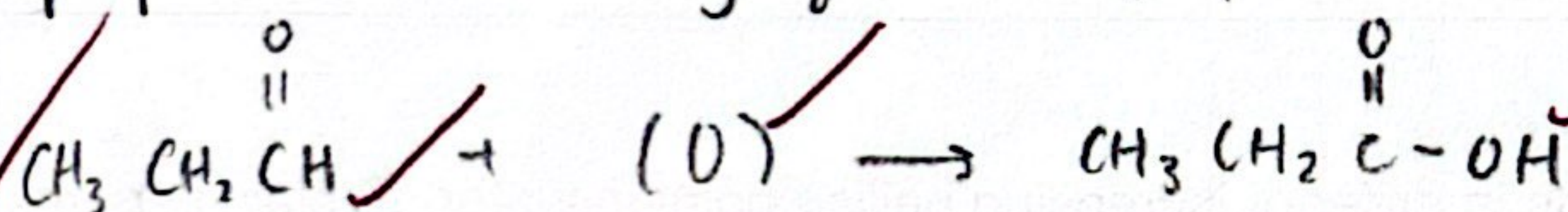
2. Complete the table below by naming the structural diagram for each of the organic compounds or drawing a structural diagram or carbon skeleton for each organic compound. (9 MARKS)

(a)  N,N-dimethyloctan-3-amine ✓	(b)  3-phenylpentanal ✓	(c)  3,5-dibromocyclopentene ✓
(d) N-methyl-N-butyl pentanamide  ✓	(e) Draw an alkyl halide with the chemical formula C₄H₈Cl₂  ✓	(f) m-chlorophenol  0.5 ✓
(g) ether  2-ethoxyhexane ✓	(h) ester  1-butyl propanoate ✓	(i)  4,5-dimethylhexanoic acid ✓

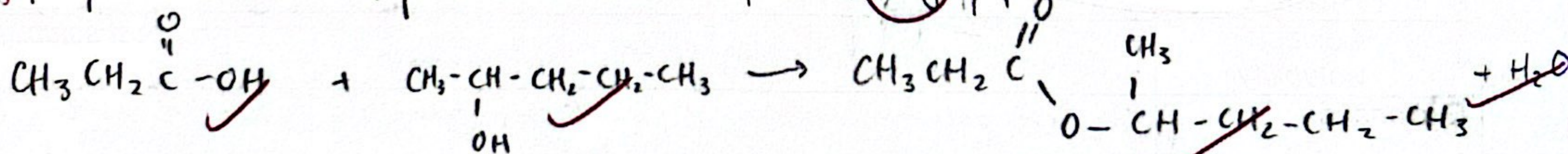
Application (8 marks)

1. You want to create an ester but only have limited supplies. Show, using a series of reactions, how you could make an ester using propanal and pentan-2-ol. Name all reactants and products and use at least 1 oxidation reaction. (5 marks)

1. propanal + oxidizing agent → propanoic acid



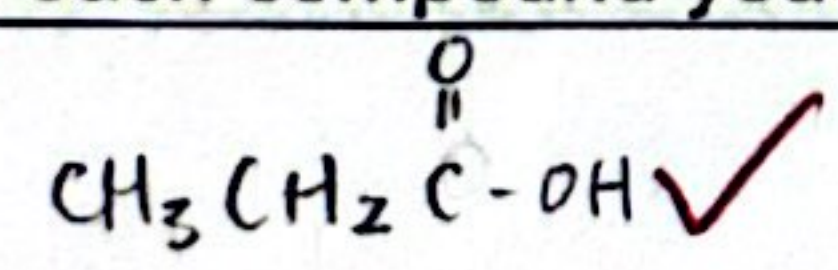
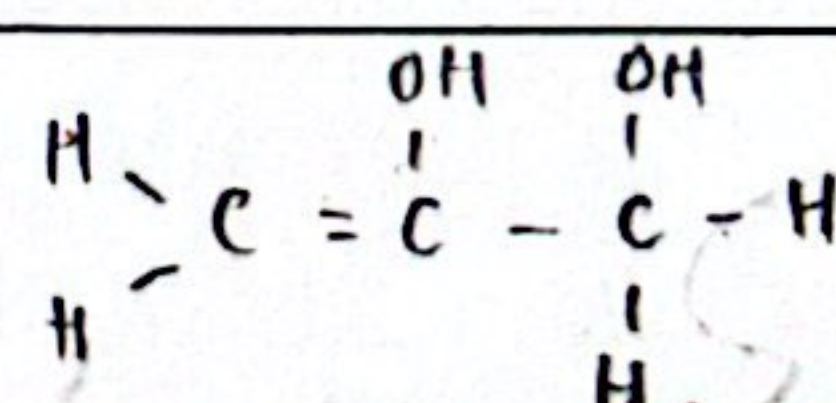
2. propanoic acid + pentan-2-ol → pentan-2-yl propanoate + water



2-pentyl propanoate

no "pentan-2-yl" in esters
for esters put number in front

2. Draw two compounds from 2 different organic families but both share the chemical formula C₃H₆O₂. Provide IUPAC name for each compound you draw. (3 marks)

Structure		
Name	propanoic acid ✓	2,3-dihydroxyprop-1-ene ✓

OR : methyl ethanoate
ethyl methanoate

love it!